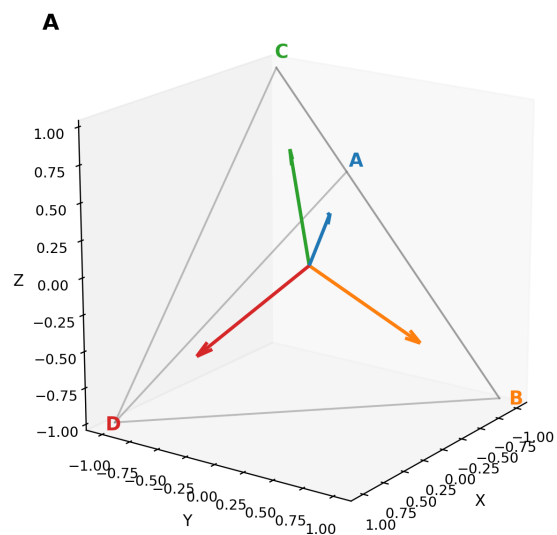
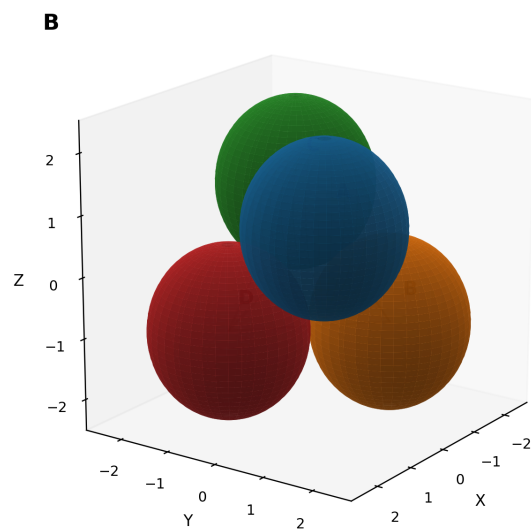


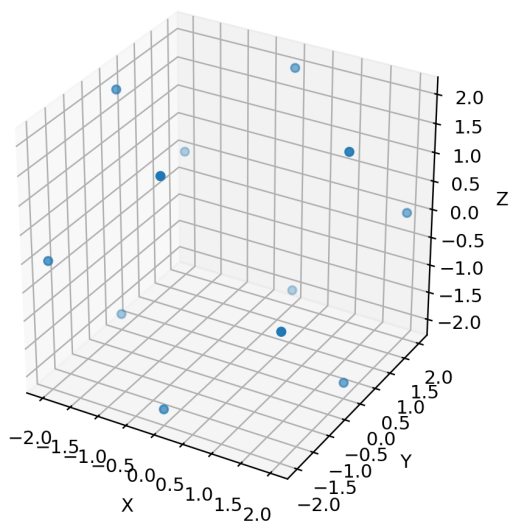
A — Quadray axes (A,B,C,D)



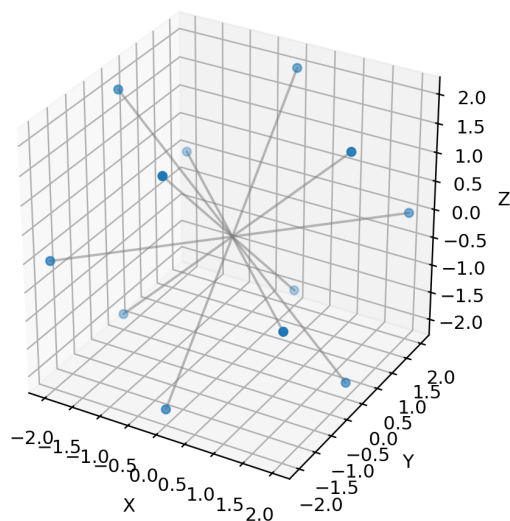
B — Tetrahedron as close-packed spheres (IVM/CCP/FCC)



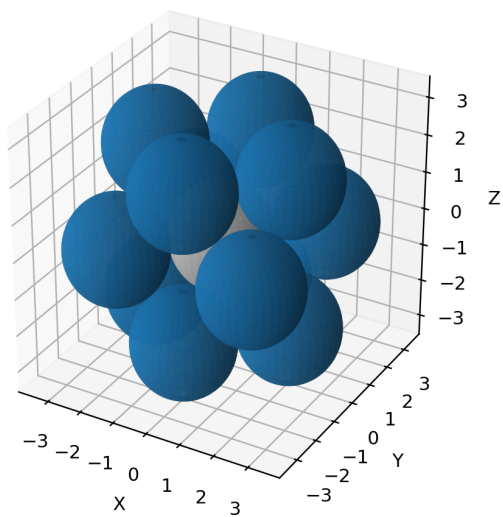
A — IVM neighbors (points)



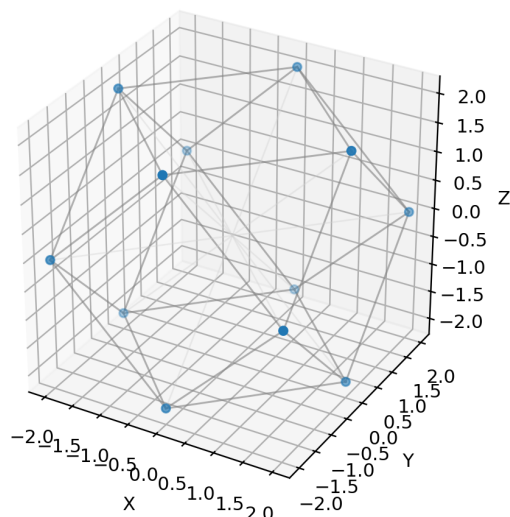
B — Neighbors with radial edges



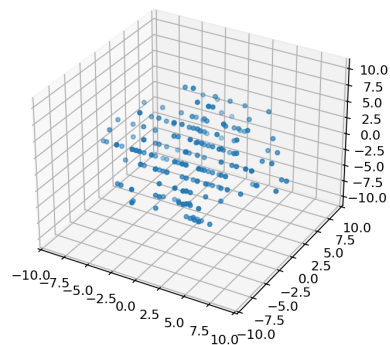
C — Twelve around one (close-packed)



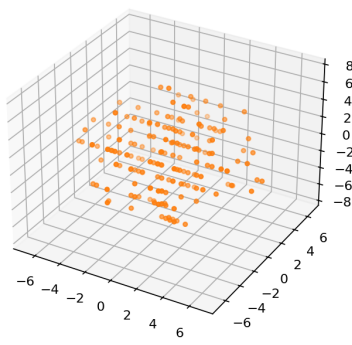
D — Vector equilibrium (adjacency)



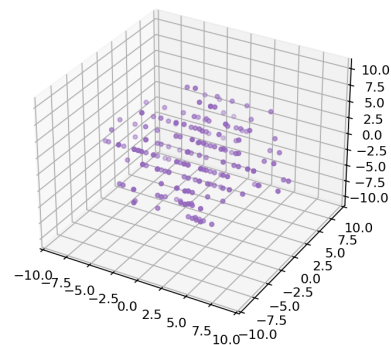
Default embedding



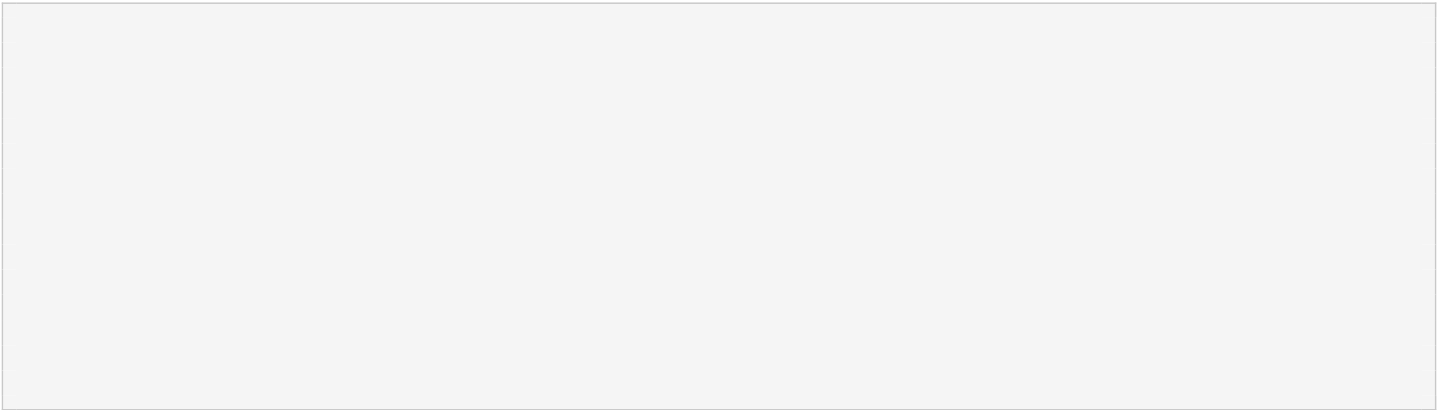
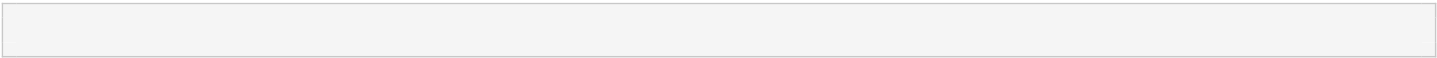
Scaled embedding (0.75x)



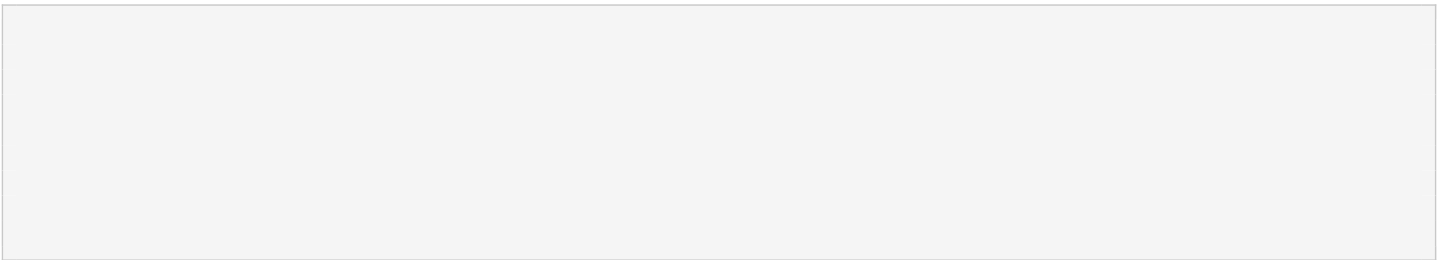
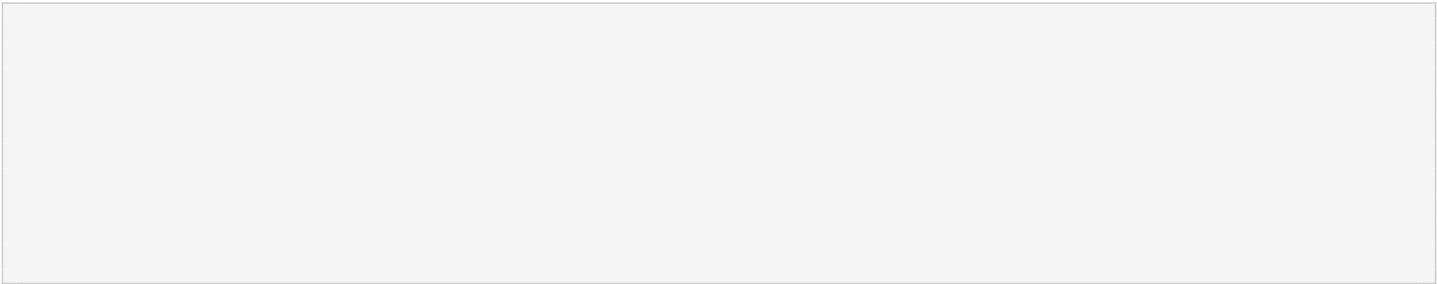
Urner embedding



$(2, 1, 1, 0)$



$(2, 1, 1, 0)$



(k, k, k, k)

$\|/4$
 $(2, 1, 1, 0)$

$(0, 0, 0, 0)(1, 0, 0, 0)(0, 1, 0, 0)(0, 0, 1, 0)\frac{1}{4}$

$$\begin{array}{l} P_0,\ldots,P_3\\ VS3=\sqrt{9/8} \end{array}$$

$$\left| \begin{array}{cc} 5 \times 5 & 3 \times 3 \end{array} \right| \left| \right| /4$$

$$\frac{1}{4}$$

$$(P_1-P_0)$$

$$d_{ij}P_iP_j$$

$$V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$P_1-P_0P_2-P_0P_3-P_0\frac{1}{4}$$

$$[i,j]=\mathbb{E}[\partial_{\boldsymbol{\theta}_i} \; p(x;\boldsymbol{\theta}) \; \partial_{\boldsymbol{\theta}_j} \; p(x;\boldsymbol{\theta})\,]$$

$$288\,V^2 = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & d_{01}^2 & d_{02}^2 & d_{03}^2 \\ 1 & d_{10}^2 & 0 & d_{12}^2 & d_{13}^2 \\ 1 & d_{20}^2 & d_{21}^2 & 0 & d_{23}^2 \\ 1 & d_{30}^2 & d_{31}^2 & d_{32}^2 & 0 \end{pmatrix}$$

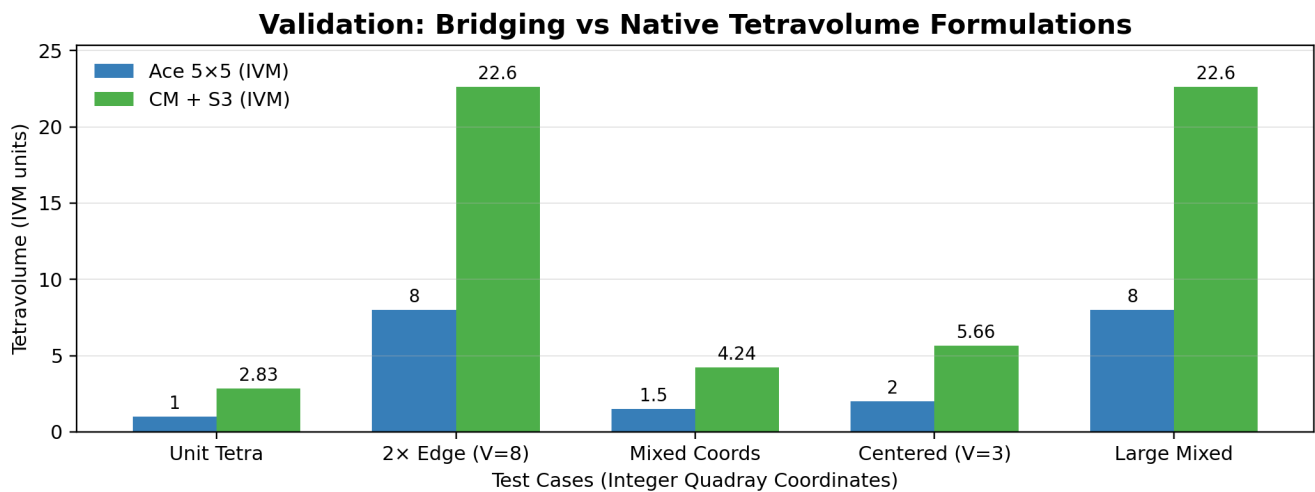
$$S3 = \sqrt{9/8}$$

$$a,b,cd,e,f$$

$$V_{ivm} = S3 \cdot V_{xyz} S3 = \sqrt{9/8}$$

$$P_1 - P_0 P_2 - P_0 P_3 - P_0 \frac{1}{4}$$

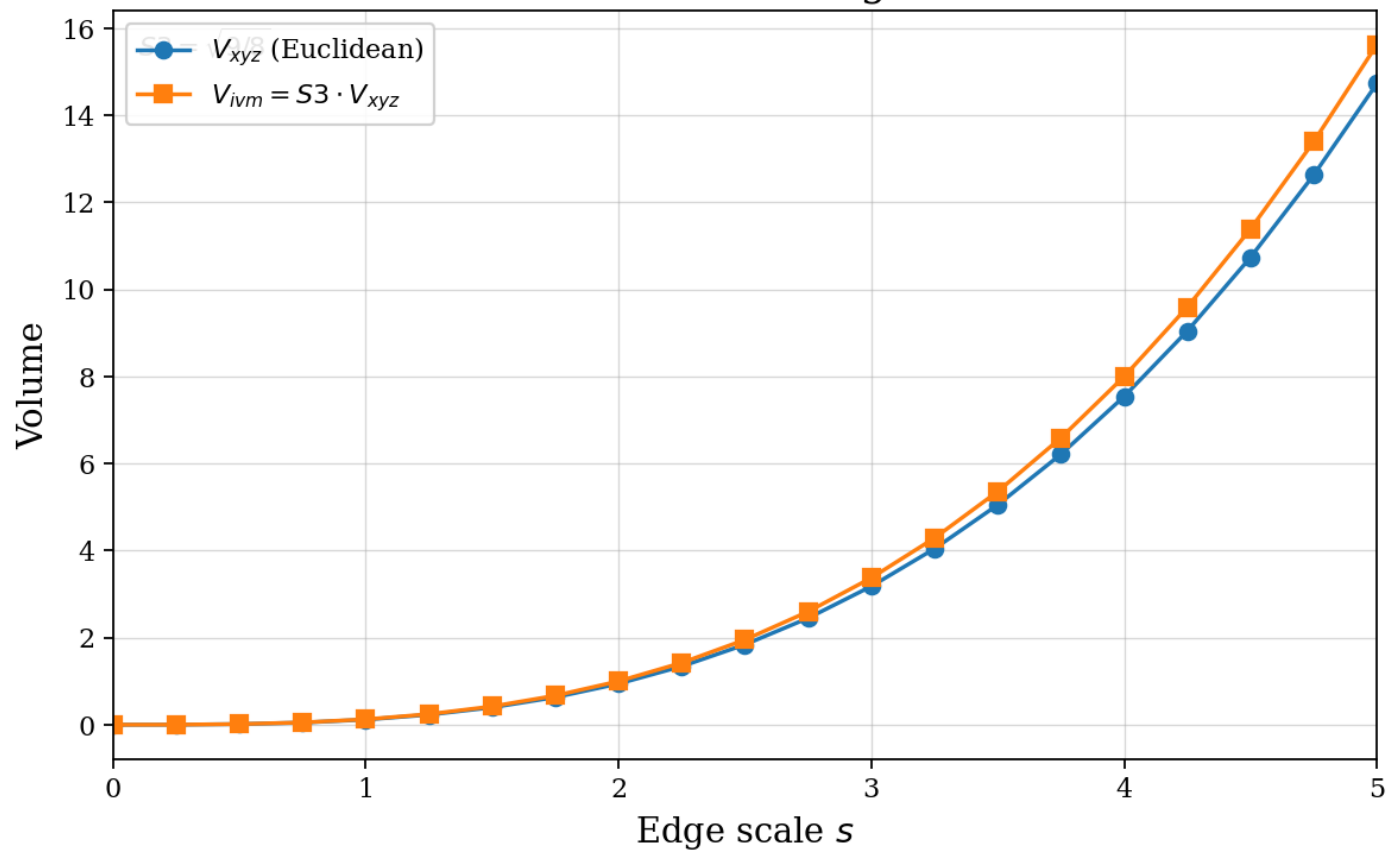
$$S3 = \sqrt{9/8}$$



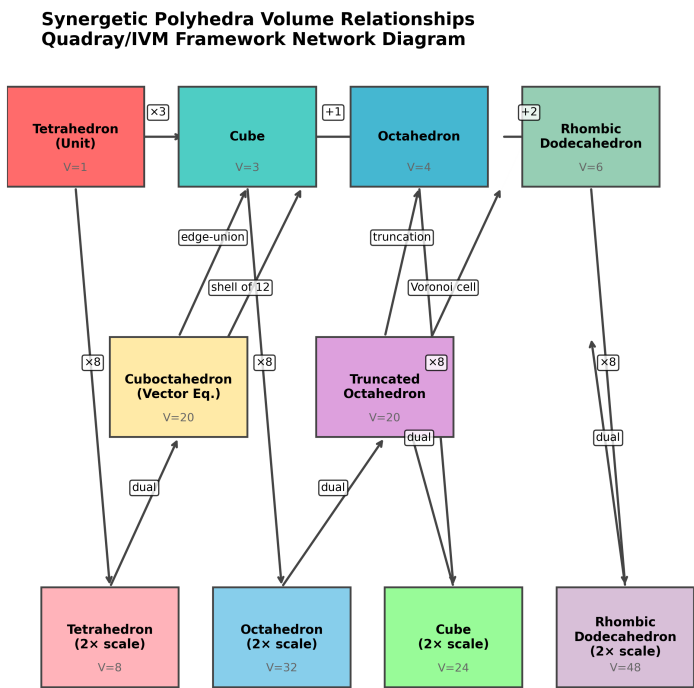
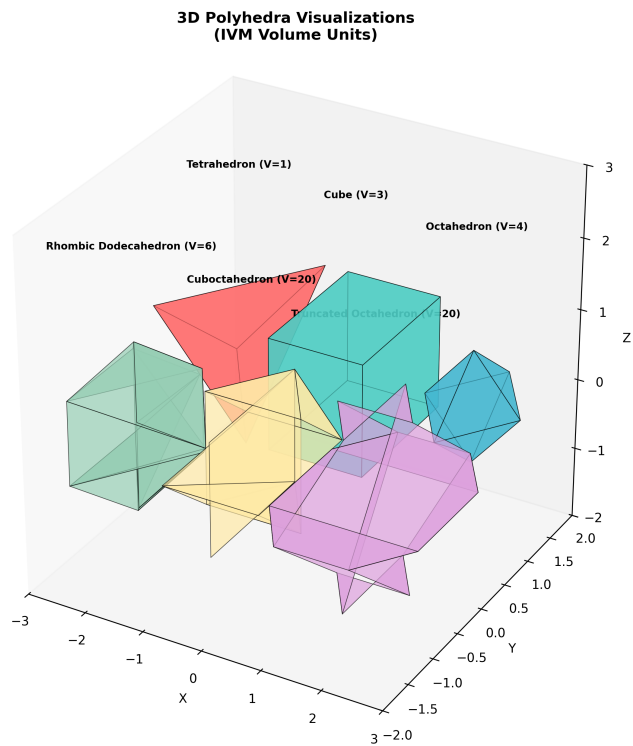
Test cases: Unit tetra (V=1), 2x edge scaling (V=8), mixed coordinates, centered (V=3), large mixed.
 Both methods agree at machine precision, validating the S3 conversion factor. Data: quadmath/output/bridging_vs_native.csv

$$S3 = \sqrt{9/8}$$

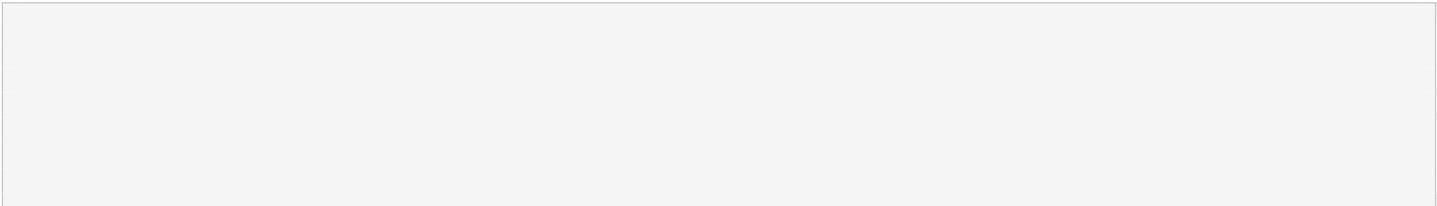
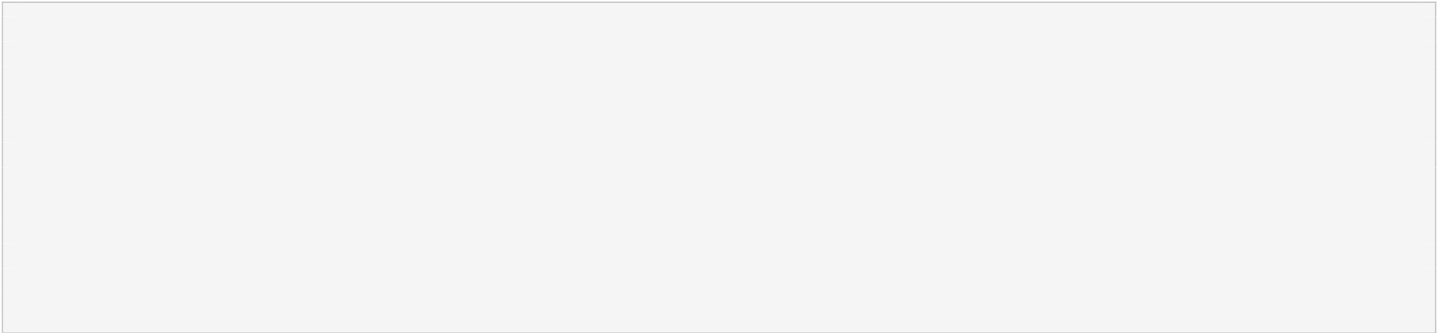
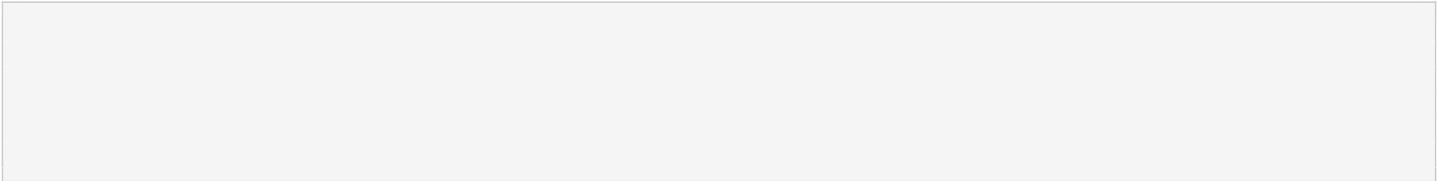
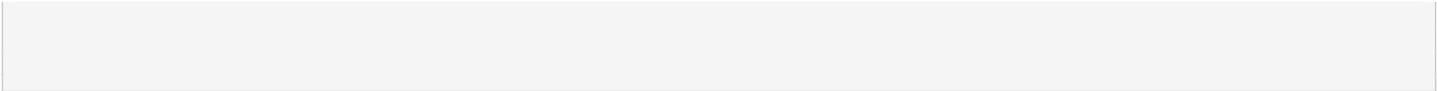
Tetra volume vs edge scale



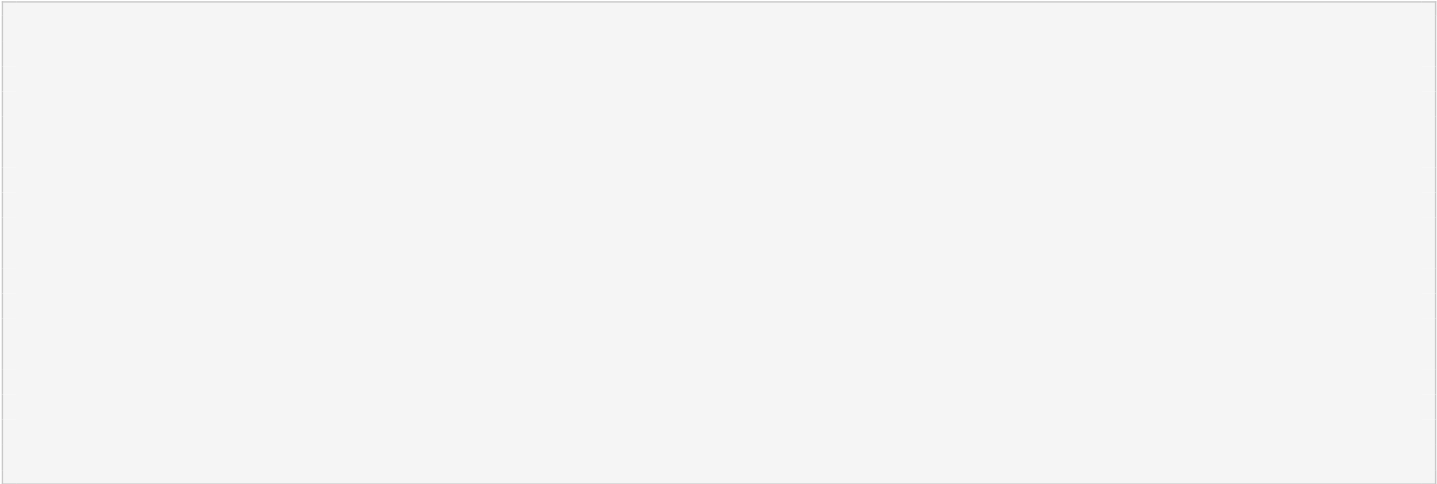
$$V = \frac{\sqrt{2}}{12} \cdot L^3 V_{ivm} = \frac{1}{8} \cdot L^3 S3 = \sqrt{9/8} \approx 1.06066$$



Volume relationships in IVM tetra-units. Regular tetrahedron (V=1) is the fundamental unit.
 Scaling relationships: 2x edge length → 8x volume ($V \propto L^3$).
 Geometric constructions: edge-union, truncation, dual relationships, and Voronoi cells.
 All polyhedra constructed with consistent edge lengths and integer Quadray coordinates.



$(2, 1, 1, 0)$

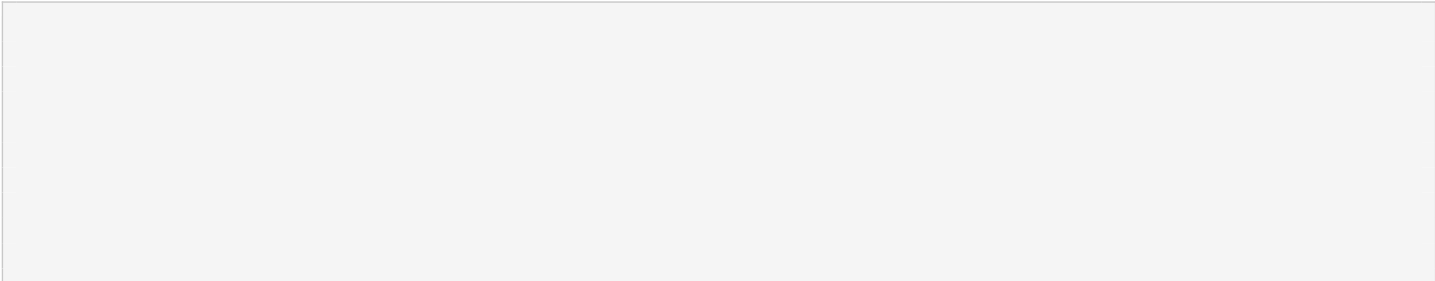
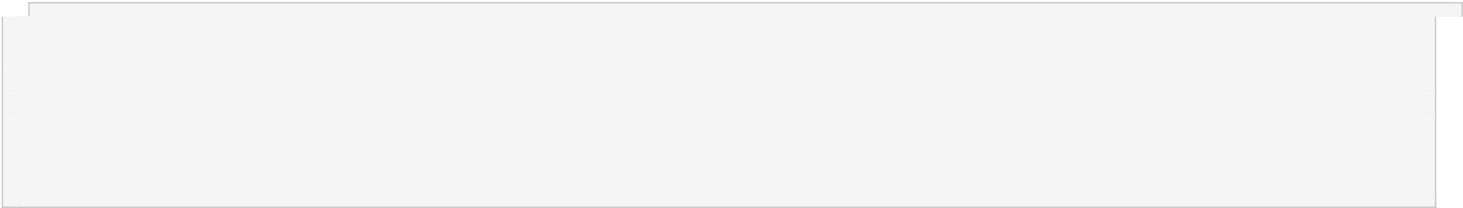


$$A,B,CABC$$

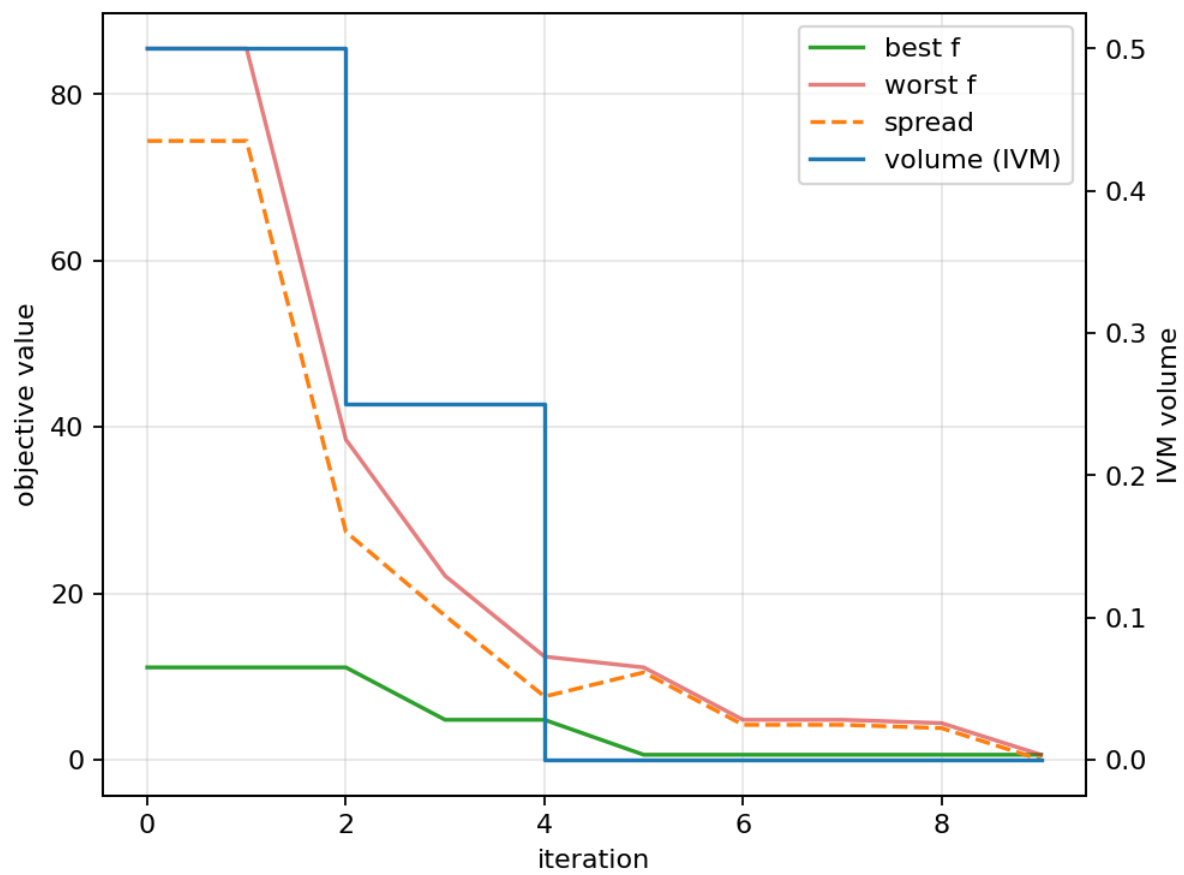
$$(P_i-P_0)$$

$$3 \times 3 \mid /4$$

$$\begin{aligned}\alpha &\approx 1 \\ \gamma &\approx 2 \\ \rho &\approx 0.5 \\ \sigma &\approx 0.5\end{aligned}$$

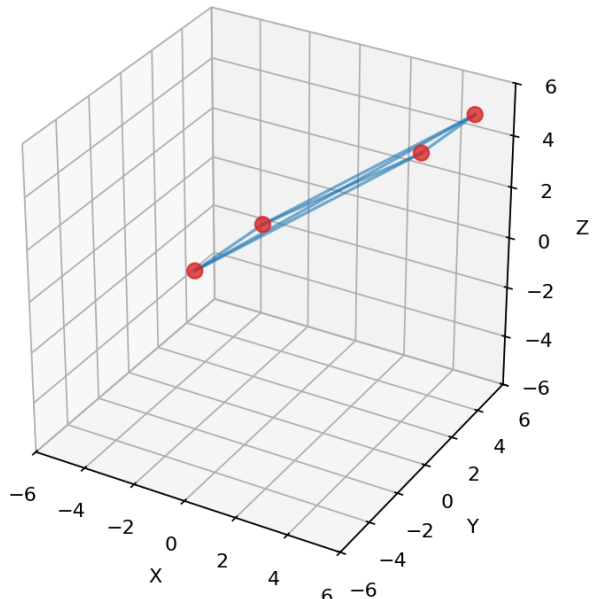


$$\begin{aligned} &F_{ij} \\ &= (w_0, w_1, w_2)L() \\ &L_n\end{aligned}$$

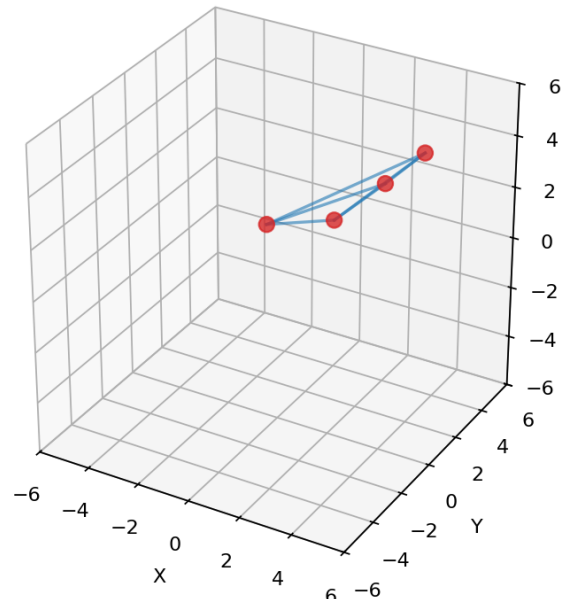


Nelder-Mead Simplex Evolution on Integer Quadray Lattice

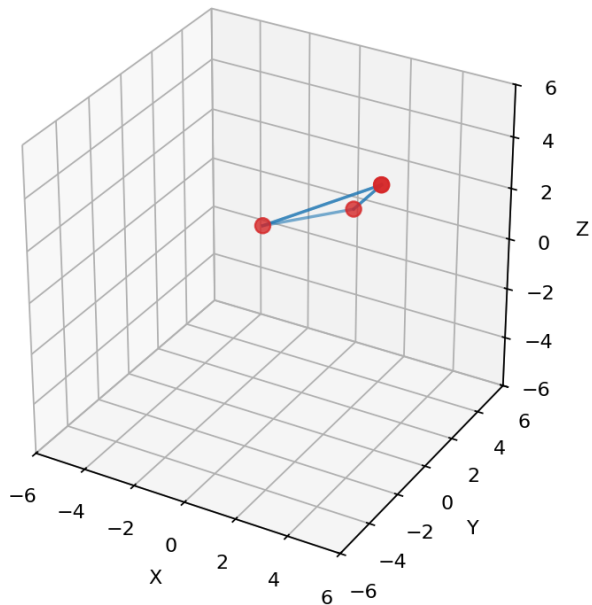
Iteration 0
Best: 11.1, Spread: 74.4



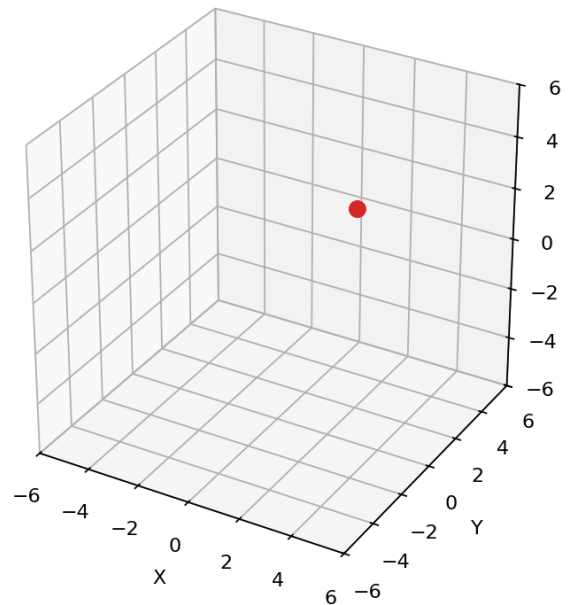
Iteration 2
Best: 4.8, Spread: 17.3



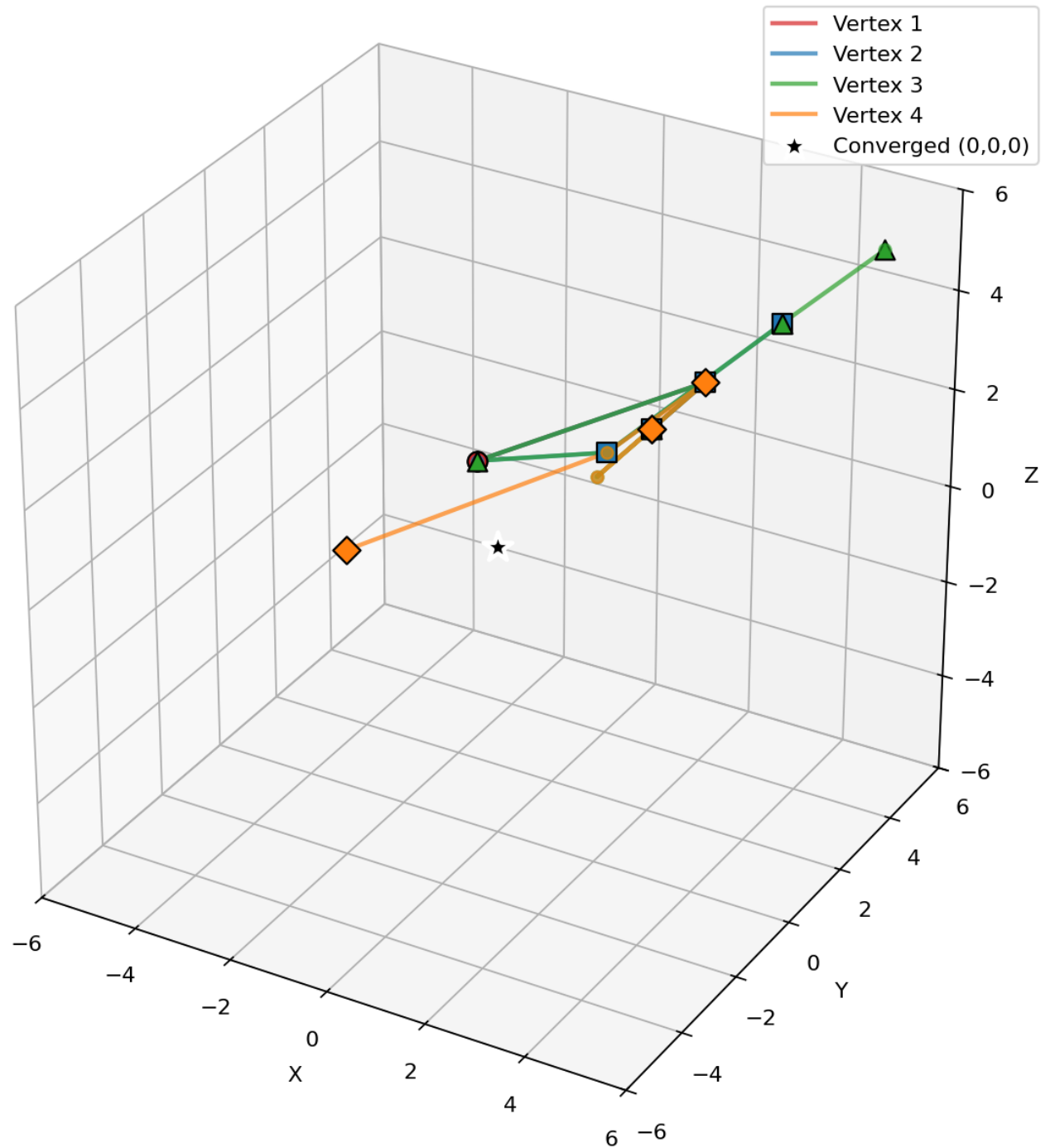
Iteration 4
Best: 0.6, Spread: 10.5



Iteration 8
Best: 0.6, Spread: 0.0



Complete Simplex Optimization Trace Nelder-Mead on Integer Quadray Lattice



$$g_{ij} = F_{ij} \delta_{ij} F^{-1} \nabla L \nabla L$$

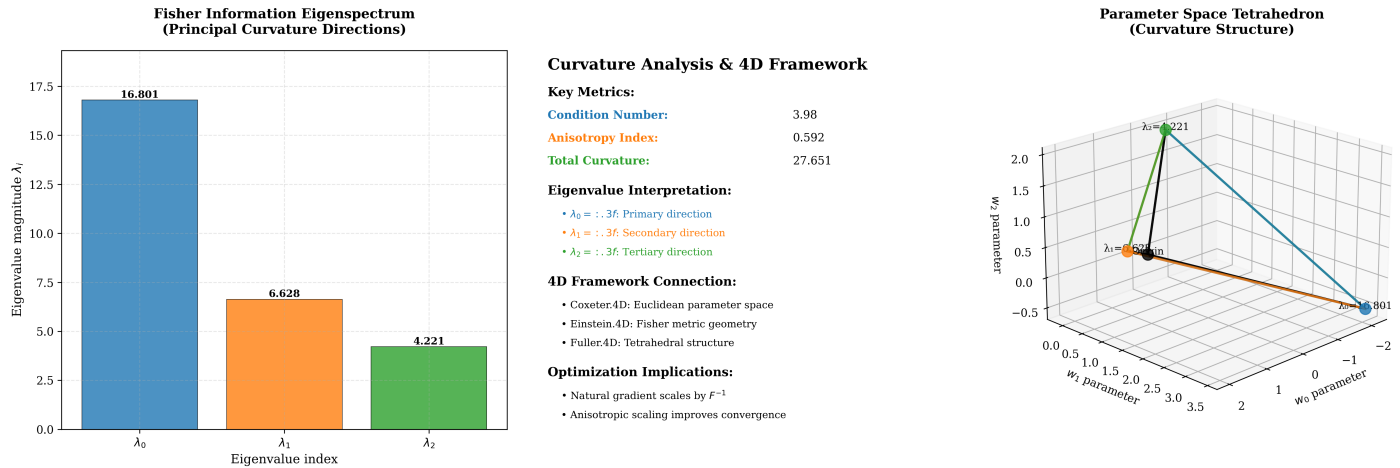
$$F^{-1} \nabla L$$

$$\delta_{ij} F_{ij}$$

$$F^{-1} \nabla L \Delta = -\eta F^{-1} \nabla L$$



$$y = w_0 + w_1x + w_2x^2w_{true} = [1.0, -2.0, 0.5]w_{est} = [0.3, -1.2, 0.0]F_{ij}F_{ii}w_iF_{ij}w_0, w_1, w_2\delta_{ij}F_{ij}\Delta w = F^{-1}\nabla L$$

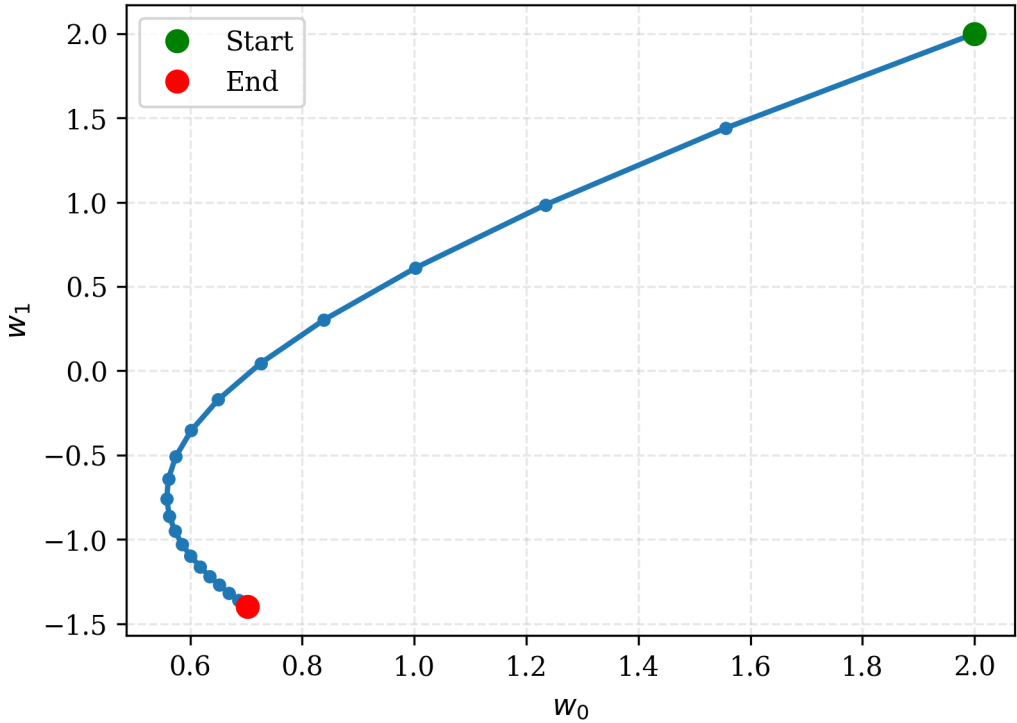


$$\lambda_i F^{-1} F^{-1}$$

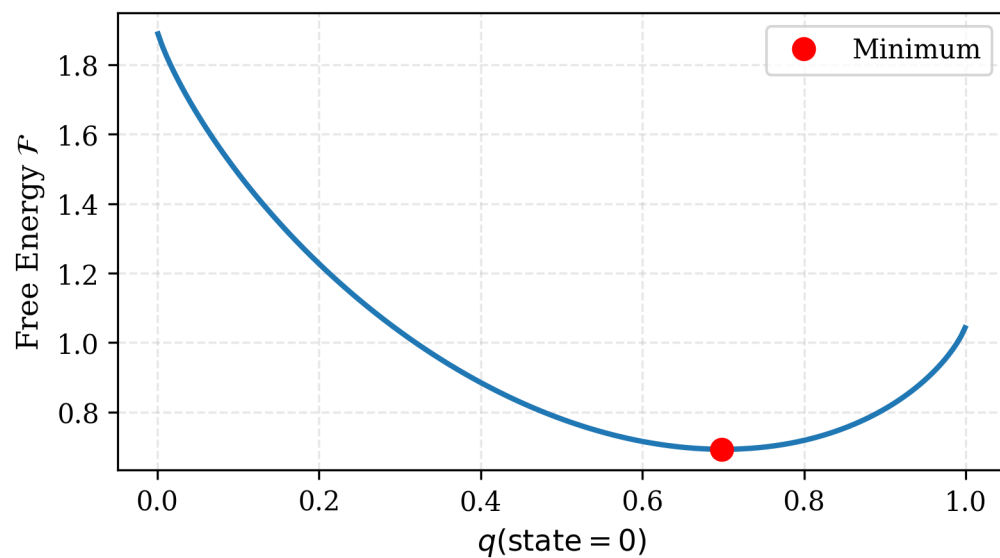
$$\eta F \nabla L g_{ij} = F_{ij}$$
$$F^{-1} \nabla L$$

F

Natural Gradient Trajectory
(Geodesic Motion on Information Manifold)



Variational Free Energy Landscape (2-State System)



$$\mathcal{F} = - P(o|s) + [Q(s)||P(s)]q(=0)\mathcal{F}$$

$$\mathcal{F} = - P(o \mid s) + [Q(s) \parallel P(s)]$$

$$V=\frac{1}{6}\parallel P_1-P_0,\;P_2-P_0,\;P_3-P_0\parallel$$

$$P_0,\ldots,P_3^{1/6}$$

$$V_{ivm}=\frac{1}{4}\left|\begin{pmatrix}a_0&a_1&a_2&a_3&1\\b_0&b_1&b_2&b_3&1\\c_0&c_1&c_2&c_3&1\\d_0&d_1&d_2&d_3&1\\1&1&1&1&0\end{pmatrix}\right|$$

$$M(q_0,q_1,q_2,q_3)=\begin{bmatrix}q_{01}&q_{02}&q_{03}&q_{04}&1\\q_{11}&q_{12}&q_{13}&q_{14}&1\\q_{21}&q_{22}&q_{23}&q_{24}&1\\q_{31}&q_{32}&q_{33}&q_{34}&1\\1&1&1&1&0\end{bmatrix},\qquad V_{ivm}=\frac{1}{4}\mid M(q_0,q_1,q_2,q_3)\mid$$

$$V_{xyz}=\frac{1}{6}\left|\begin{pmatrix}x_a&y_a&z_a&1\\x_b&y_b&z_b&1\\x_c&y_c&z_c&1\\x_d&y_d&z_d&1\end{pmatrix}\right|,\qquad V_{ivm}=S3\,V_{xyz},\quad S3=\sqrt{\frac{9}{8}}$$

$$S3=\sqrt{9/8}$$

$$_{288}V^2=\begin{pmatrix}0&1&1&1&1\\1&0&d_{01}^2&d_{02}^2&d_{03}^2\\1&d_{10}^2&0&d_{12}^2&d_{13}^2\\1&d_{20}^2&d_{21}^2&0&d_{23}^2\\1&d_{30}^2&d_{31}^2&d_{32}^2&0\end{pmatrix}$$

$$d_{ij}P_iP_j$$

$$V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8}$$

$$144\,V_{xyz}^2=4a^2b^2c^2-a^2\,(b^2+c^2-f^2)^2-b^2\,(c^2+a^2-e^2)^2-c^2\,(a^2+b^2-d^2)^2+(b^2+c^2-f^2)(c^2+a^2-e^2)(a^2+b^2-d^2)$$

$$\begin{array}{c} a,b,cd,e,f \\ V_{ivm}=S3\cdot V_{xyz}S3=\sqrt{9/8} \end{array}$$

$$V_{ivm}=\frac{1}{4}\left|\begin{pmatrix}a_1-a_0&a_2-a_0&a_3-a_0\\b_1-b_0&b_2-b_0&b_3-b_0\\c_1-c_0&c_2-c_0&c_3-c_0\end{pmatrix}\right|$$

$$P_1-P_0P_2-P_0P_3-P_0$$

$$F_{i,j} = \mathbb{E}\left[\frac{\partial}{\partial \theta_i} p(x;\theta) \frac{\partial}{\partial \theta_j} p(x;\theta)\right]$$

$$F_{i,j} = \frac{1}{N} \sum_{n=1}^N \frac{\partial}{\partial \theta_i} p(x_n;\theta) \frac{\partial}{\partial \theta_j} p(x_n;\theta)$$

$$NN\rightarrow\infty$$

$$\theta \leftarrow \theta - \eta \, F(\theta)^{-1} \, \nabla_{\theta} L(\theta)$$

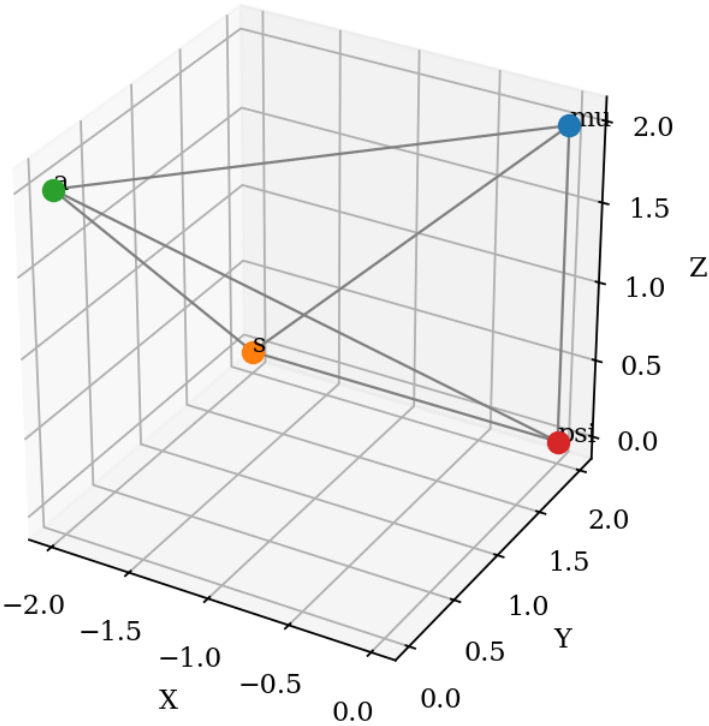
$$\mathcal{F} = -\, P(o \mid s) + [Q(s) \parallel P(s)]$$

$$q=(a,b,c,d)k=(a,b,c,d)q'=q-(k,k,k,k)$$

$$M\mathbb{R}^3\mathbb{R}^4d(q_1,q_2)=\|M(q_1)-M(q_2)\|_2$$

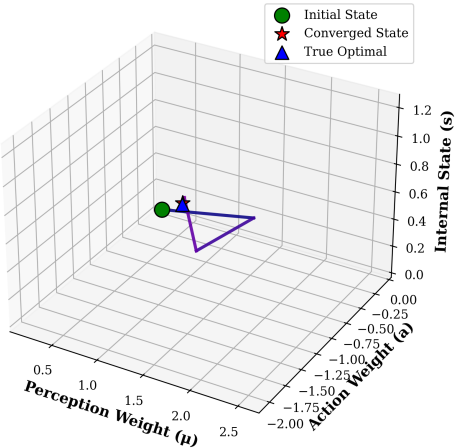
$$ds^2=-c^2\,dt^2+dx^2+dy^2+dz^2$$

Four-fold partition mapped to Quadray tetrahedron

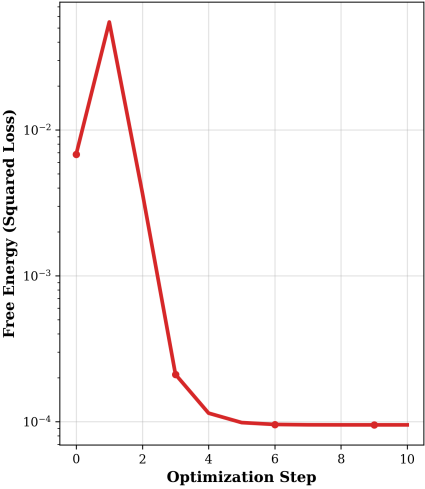


μ
 s
 a
 ψ

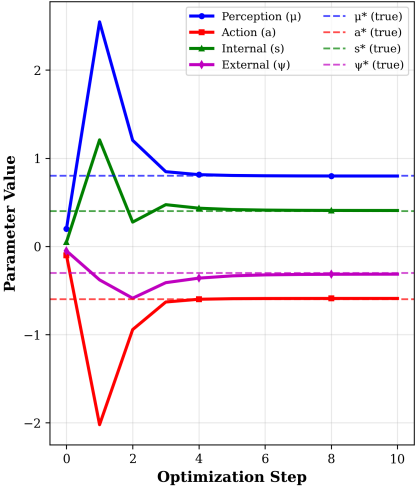
4D Natural Gradient Trajectory
(Active Inference Dynamics)



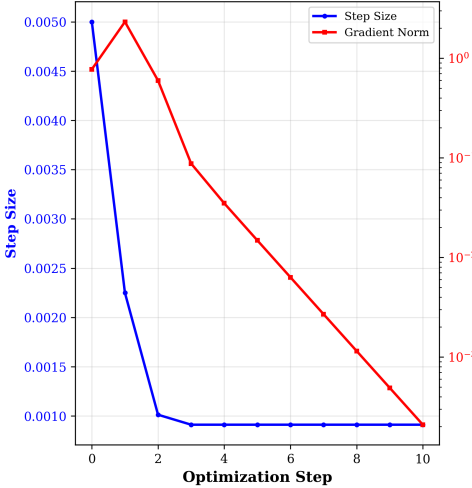
Free Energy Minimization
(Active Inference Principle)



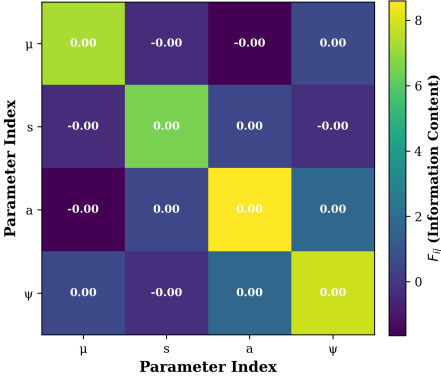
Four-Fold Partition Evolution
(Fuller.4D Tetrahedral Structure)



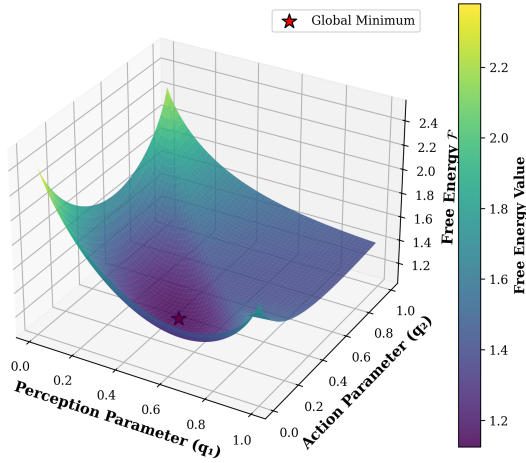
Optimization Diagnostics
(Step Size & Gradient Evolution)



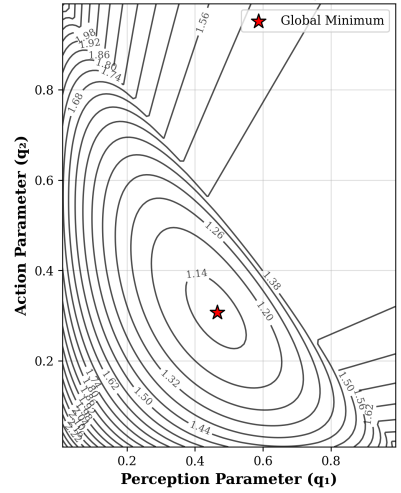
Fisher Information Matrix
(Information Geometry Metric)



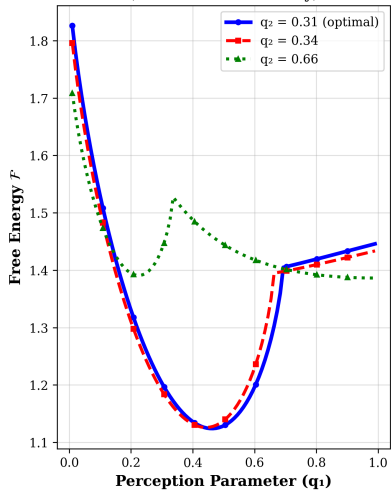
Free Energy Landscape
(4D Active Inference Framework)



Free Energy Contours
(Information Geometry)



Free Energy Cross-Sections
(Parameter Sensitivity)



Active Inference Four-Fold Partition:

μ (Internal States)
 s (Sensory) a (Actions)
 ψ (External Causes)

4D Framework Integration:

- Fuller.4D: Tetrahedral structure with IVM quantization
- Coxeter.4D: Exact Euclidean geometry for measurements
- Einstein.4D: Information-geometric flows and geodesics

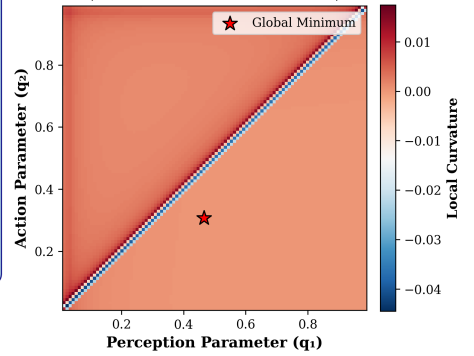
Free Energy Principle:

- Minimization drives perception-action cycles
- Balances prediction error vs. complexity
- Natural gradient follows geodesics on information manifold

Mathematical Structure:

- Variational posterior $Q(s)$ over latent states
- Prior $P(s)$ encoding environmental structure
- Likelihood $P(o|s)$ connecting observations to states

Local Curvature
(Fisher Information Structure)



$$\frac{\mu s \nabla}{a \psi \nabla_a} \mu^F$$

$$\mathbb{R}^n$$

$$\mathbb{R}^{n_n}$$

$$\begin{array}{l} q=(a,b,c,d)\\ A,B,C,D\\ k\\ q'\\ P_0,\ldots,P_3\\ d_{ij}\\ (\cdot)\\ |\cdot|\\ V_{ivm}\\ V_{xyz}\\ S3 \end{array}$$

$$k=(a,b,c,d)q'=q-(k,k,k,k)$$

$$P_iP_j$$

$$V_{ivm}=1$$

$$S3=\sqrt{9/8}V_{ivm}=S3\,V_{xyz}$$

$$\begin{array}{l} \alpha\\ \gamma\\ \rho\\ \sigma\\ V_{ivm} \end{array}$$

$$\begin{array}{l} \alpha,\gamma,\rho,\sigma\\ V_{ivm} \end{array}$$

$$e$$

$$\mathbb{E}[\cdot]$$

$$F_{ij}$$

$$\mathcal{F}$$

$$[Q\|P]$$

$$\nabla_{\theta}L$$

$$\eta$$

$$\theta$$

$$ds^2$$

$$c$$

$$\mathbb{E}[\partial_{\theta_i}\,p\cdot\partial_{\theta_j}\,p]$$

$$-P(o\mid s)+[Q(s)\parallel P(s)]$$

$$\sum Q\left(Q/P\right)$$

$$F(\theta)^{-1}\nabla_{\theta}L(\theta)$$

$$\theta_i$$

$$-c^2\,dt^2+dx^2+dy^2+dz^2$$

$$M$$

$$\|\cdot\|_2$$

$$R,D$$

$$\mathbb{R}^3$$

$$\sqrt{x_1^2+\cdots+x_n^2}$$

$$RDD=2R$$

$$\alpha,\gamma,\rho,\sigma$$

$$\theta$$

$$\mu$$

$$\psi$$

$$\eta$$